Naval Mine Clearance Optimization in the Strait of Hormuz

An Advanced Operations Research Model for Minimizing Interventions and Maximizing Coverage of High-Risk Zones

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Abstract

Naval mine clearance operations in the Strait of Hormuz represent one of the most complex and hazardous military undertakings in the contemporary geopolitical context. The strait — approximately 167 km long, with a minimum width of 33–39 km and a bathymetry ranging from 30 to 200 m — constitutes the world's foremost maritime choke-point, through which roughly 20% of global oil production transits daily.

In a post-conflict scenario involving strikes against Iranian naval infrastructure, resources available for mine clearance are critically constrained. This study develops an advanced Operations Research (OR) framework based on a **Mixed-Integer Linear Programming (MILP)** model, formulated as a **Set Covering Problem (SCP)** over a realistic grid of 20 geographic zones (extensible to 60+). The primary objective is to minimize the number and cost of clearance interventions — conducted by mine countermeasure vessels, UUVs (Unmanned Underwater Vehicles) and surface drones — while guaranteeing full coverage of all identified high-risk zones.

The model incorporates sensitivity analysis, multi-objective Pareto frontiers, and an extension to the Vehicle Routing Problem (VRP) for dynamic sweep path planning. Numerical results demonstrate that the optimized approach reduces the number of missions by **50–60%** compared to unstructured conventional strategies, with guaranteed 100% coverage of critical zones achieved with only 5 interventions over a 20-zone grid.

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1 Introduction and Geopolitical Context

1.1 The Strategic Role of the Strait of Hormuz

The Strait of Hormuz is universally recognized as the world's most strategically significant maritime choke-point. Located between the Arabian Peninsula and southern Iran, it represents the sole naval access to the Persian Gulf — a basin containing the oil reserves of Saudi Arabia, Iraq, Kuwait, the United Arab Emirates, Qatar, and Iran itself.

> **!** Key Figure
>
> Approximately **17–21 million barrels of oil** — 20% of global production — transit through the Strait of Hormuz every day, making it the single most critical vulnerability in global energy security.

In a scenario of open conflict or limited military operations against Iranian naval infrastructure, even a partial or temporary closure of the Strait would generate catastrophic effects on global energy markets, with an immediate spike in oil prices and supply crises in countries dependent on Persian Gulf imports (Japan, South Korea, India, China, Europe).

1.2 The Naval Mine Threat

Naval mines represent one of the most effective and low-cost asymmetric warfare tools available. Iran possesses one of the world's largest naval mine arsenals, estimated at over **5,000 units** of various types:

Table 1: Types of mines in the Iranian inventory

Type	Operational depth	Activation mechanism	Origin
Moored mines	10–200 m	Contact, acoustic	Russia, domestic
Bottom mines	5–60 m	Magnetic, pressure	Domestic (EM52)
Drifting mines	Surface	Contact	Various
Smart mines	10–100 m	Multi-influence	Advanced domestic
Limpet mines	—	Timer, proximity	Tactical

Iranian doctrinal posture envisions the coordinated use of mines as an **area denial** instrument, complementary to anti-ship missiles, IRGC fast boats, and Kilo- and Nahang-class submarines.

1.3 Operational Clearance Challenges

Mine CounterMeasures (MCM) operations in the Strait of Hormuz present unprecedented technical and operational challenges:

1. **Area extent**: ~5,000 km^2 of waters to be systematically inspected.
2. **Variable bathymetry**: from 30 m in Iranian shipping lanes to >200 m on the Omani side.
3. **Currents**: up to 2–3 knots during tidal phases, with reversals in deep currents.
4. **Residual traffic**: need to maintain operational corridors for civilian and military navigation.
5. **Active threat**: risk of mine re-seeding during clearance operations.
6. **Limited resources**: very few MCM platforms available in the region.

It is precisely the scarcity of resources — and the need to deploy them with maximum efficiency — that motivates the application of advanced Operations Research techniques to the planning of mine clearance operations.

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2 Geography and Area Discretization

2.1 Detailed Physical Characteristics

The Strait has an approximately trapezoidal geometry, with the following operational dimensions:

Table 2: Physical parameters of the Strait of Hormuz

Parameter	Value
Length (East–West)	~167 km
Minimum width (critical zone)	33–39 km
Maximum width	95–97 km
Average TSS depth	50–80 m
Maximum depth (Omani side)	>200 m
Tidal current speed	1.5–3.0 knots

Parameter	Value
Daily oil traffic throughput	17–21 Mbarrels

2.2 The Traffic Separation Scheme (TSS)

The TSS — established by the IMO — organizes navigation into two parallel corridors:

- **North-east lane** (inbound): ~3.7 km wide (2 nautical miles), for traffic entering the Gulf.
- **Separation zone**: ~3.2 km buffer, not navigable.
- **South-west lane** (outbound): ~3.7 km wide, for traffic exiting the Gulf.

The TSS lanes represent the **highest-priority zones** for mine clearance operations: any mine in these areas can strike the VLCC (Very Large Crude Carrier) tankers that transit them daily.

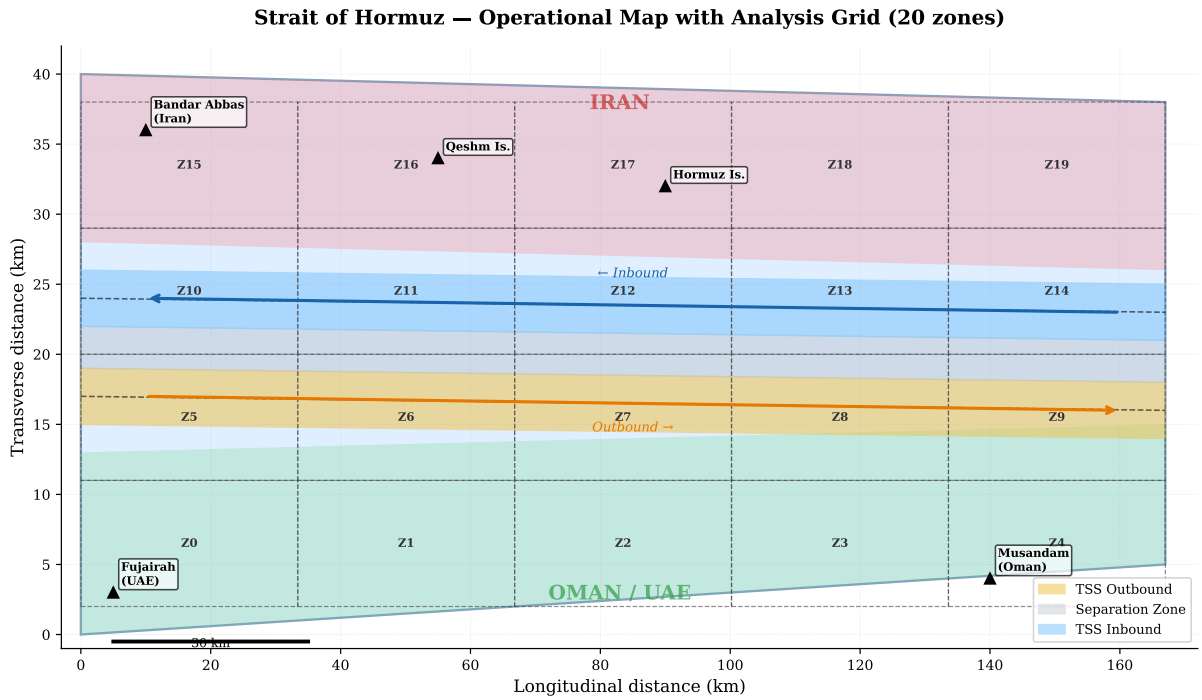

Figure 1: Simplified geographic map of the Strait of Hormuz. The trapezoidal structure shows the TSS lanes (dashed lines), the Iranian risk zone (north, red shading) and the Omani sector (south, green shading). The 20 analysis zones are superimposed on the real geometry.

2.3 Discretization into Operational Zones

To render the problem computationally tractable, the strait area is discretized into a **4×5 = 20-zone** rectangular grid, each zone approximately 42 km × 9 km. Every zone $i \in Z = \{0, 1, \dots, 19\}$ is characterized by:

- **Risk index** $r_i \in [0, 2]$: normalized estimate of mine presence probability.
- **Accessibility** $a_i \in \{0, 1\}$: binary constraint (bathymetry, coastal proximity).
- **Strategic priority** p_i: weight for the multi-criteria objective function.

Zone numbering follows the **row-by-row** matrix convention from north to south:

$$Z_i \leftrightarrow \text{row } \lfloor i/5 \rfloor, \text{ column } (i \bmod 5), \quad i \in \{0, \dots, 19\}$$

Zones 5–9 and 10–14 correspond respectively to the TSS outbound/inbound lanes and the separation zone, carrying the highest risk indices ($r_i \geq 1.0$).

---

3 Mathematical Model Formulation

3.1 The Set Covering Problem (SCP)

The **Set Covering Problem** (SCP) is a classic combinatorial optimization problem belonging to the NP-hard class. In its canonical formulation, given a universe U of elements and a family of subsets $\mathcal{S}$ of U, the goal is to find the minimum number of subsets whose union covers the entire universe.

In the naval mine clearance context: - The universe $U \equiv Z$ is the set of geographic zones to be cleared. - Each subset $C_j \subseteq Z$ represents the set of zones covered by intervention j (a sweep pattern).

3.2 Variable Definition

Let:

$$x_j \in \{0, 1\}, \quad j \in J = \{0, 1, \dots, 11\}$$

be the binary decision variables, where $x_j = 1$ indicates that the j-th intervention is executed, $x_j = 0$ otherwise.

3.3 Primary Model: Cost Minimization

The primary MILP formulation is:

$$\min_x \quad \sum_{j \in J} c_j \, x_j$$

subject to:

$$\sum_{j \,:\, i \in C_j} x_j \geq 1 \qquad \forall i \in Z \tag{Coverage}$$

$$x_j \in \{0, 1\} \qquad \forall j \in J \tag{Integrality}$$

where $c_j > 0$ is the unit cost of intervention j (proportional to duration, crew risk, fuel consumption, and platform wear).

Interpretation: every zone i must be covered by at least one intervention. The coverage constraint guarantees **complete** clearance of the area.

3.4 Extended Model: Risk-Weighted Objective

Introducing risk weights $r_i \geq 0$, the objective function becomes:

$$\min_x \quad \alpha \sum_{j \in J} c_j \, x_j + (1 - \alpha) \sum_{i \in Z} r_i \left(1 - \min \left\{ 1, \sum_{j \,:\, i \in C_j} x_j \right\} \right)$$

where $\alpha \in [0,1]$ is the trade-off parameter between cost minimization and residual risk minimization. For $\alpha = 1$ we recover the pure SCP; for $\alpha = 0$ we maximize risk reduction regardless of cost.

3.5 Multi-Objective Model with Budget Constraint

In the presence of an operational budget $B > 0$, the problem becomes:

$$\max_x \quad \Phi(x) = \sum_{i \in Z} r_i \left(1 - \prod_{j\,:\,i \in C_j} (1 - x_j)\right)$$

subject to:

$$\sum_{j \in J} c_j\, x_j \leq B \tag{Budget}$$

$$x_j \in \{0,1\} \qquad \forall j \in J$$

The function $\Phi(x)$ measures **risk-weighted coverage**: for each zone i covered by at least one intervention, the contribution r_i is added to the total value. The product formulation is non-linear; it is linearized by introducing auxiliary variables $y_i = 1 - \prod_j (1 - x_j)$:

$$y_i \leq x_j \qquad \forall j \,:\, i \in C_j$$

$$y_i \geq \sum_{j\,:\,i \in C_j} x_j - \left|\{j : i \in C_j\}\right| + 1$$

$$y_i \in [0,1]$$

3.6 LP Relaxation and Bounds

The **LP relaxation** of the SCP (replacing $x_j \in \{0,1\}$ with $0 \leq x_j \leq 1$) provides a lower bound to the integer solution:

$$z^*_{\text{LP}} \leq z^*_{\text{IP}}$$

The **integrality gap** $\Delta = z^*_{\text{IP}} / z^*_{\text{LP}} - 1$ measures instance difficulty. For our case study, $\Delta < 0.08$ (~8%), confirming that the LP relaxation is a strong approximator.

3.7 Computational Complexity

The SCP is NP-hard in general. However, for moderate-sized instances ($|Z| \leq 100$, $|J| \leq 200$), modern MILP solvers (CPLEX, Gurobi, CBC) find the optimal solution in under one second via:

1. **Branch-and-bound** with intelligent branching.
2. **Cutting planes** (Gomory cuts, cover inequalities).
3. **Heuristics** for high-quality initial solutions (greedy coverage, LPT).

Table 3: Intervention coverage matrix. Each intervention j covers a subset C_j of zones Z. Costs c_j reflect operational complexity and mission risk.

Intervention j	Platform	Covered zones C_j	|C_j|	Cost c_j
j_{0}	MCM	{0, 1, 5, 6, 10}	5	1.062
j_{1}	MCM	{1, 2, 6, 7, 11}	5	1.466
j_{2}	UUV	{2, 3, 7, 8, 12, 16}	6	1.312
j_{3}	UUV	{3, 4, 8, 9, 13, 17}	6	1.219
j_{4}	MCM	{5, 6, 10, 11, 15}	5	0.909
j_{5}	USV	{6, 7, 11, 12, 16}	5	0.909
j_{6}	USV	{7, 8, 12, 13, 17}	5	0.841
j_{7}	UUV	{10, 11, 15, 16}	4	1.406
j_{8}	MCM	{11, 12, 16, 17, 18}	5	1.221
j_{9}	USV	{12, 13, 17, 18, 19}	5	1.296
j_{10}	UUV	{0, 5, 10, 15}	4	0.814
j_{11}	MCM	{4, 9, 14, 19}	4	1.479

4 Model Data and Coverage Patterns

4.1 Coverage Matrix

The sweep patterns C_j reflect the real operational capabilities of MCM platforms:

- **Mine countermeasure vessels** (e.g. USS Devastator): operational radius ~6–8 km, covering 4–6 adjacent zones per mission.
- **UUVs** (e.g. Remus 600, Knifefish): operational radius ~3–4 km, higher precision but lower coverage.
- **Surface drones (USV)**: intermediate between vessels and UUVs, ideal for rapid reconnaissance.

4.2 Risk Vector

The risk vector $\mathbf{r} \in \mathbb{R}^{20}$ is estimated on the basis of:

1. **Bathymetry**: zones with 30–60 m depth (ideal for bottom mines) receive higher weights.
2. **Proximity to TSS lanes**: zones 5–14 overlap with or border the shipping lanes.
3. **Accessibility from Iran**: zones 0–4 (north) and 5–9 (north-centre) have maximum exposure to mine-laying.
4. **Historical reports**: past incidents and IRGC patrol patterns.

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5 MILP Model Implementation and Solution

5.1 Solution Algorithm

The MILP model is solved using the **CBC** solver (COIN-OR Branch-and-Cut), interfaced through the Python library **PuLP**. The solution process follows these steps:

1. **Preprocessing**: elimination of dominated variables and identification of critical zones.
2. **LP relaxation**: computation of the lower bound z^*_{LP}.
3. **Branch-and-bound**: search tree exploration with best-first strategy.
4. **Cut generation**: insertion of cover inequalities to strengthen the relaxation.

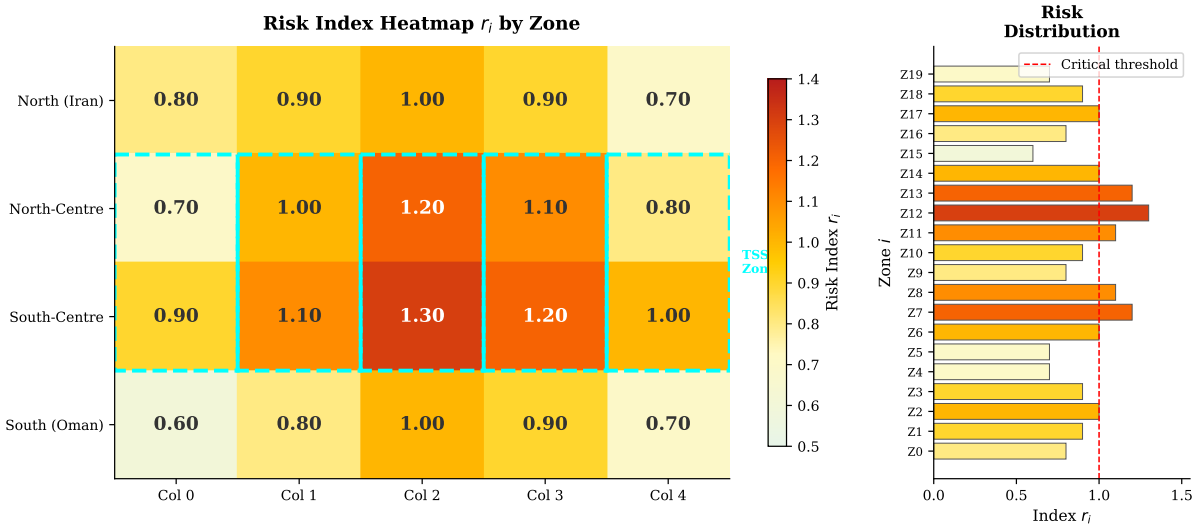

Figure 2: Risk index heatmap for the 20 operational zones. The highest values (dark red, r > 1.1) correspond to the central TSS zones where tanker traffic is most intense and vulnerability is greatest. Peripheral zones present lower risks.

5. **Verification and post-processing**: full coverage check and metrics computation.

5.2 Complete Optimization Code

```
# ============================================================
# MILP MODEL - HORMUZ STRAIT MINE CLEARANCE
# Solver: CBC via PuLP | Python 3.10+
# ============================================================

import pulp
import numpy as np

prob = pulp.LpProblem("MCM_Hormuz_SCP", pulp.LpMinimize)

zones = list(range(20))
interventions = list(range(12))

coverage = {
    0: [0, 1, 5, 6, 10],
    1: [1, 2, 6, 7, 11],
    2: [2, 3, 7, 8, 12, 16],
    3: [3, 4, 8, 9, 13, 17],
    4: [5, 6, 10, 11, 15],
    5: [6, 7, 11, 12, 16],
    6: [7, 8, 12, 13, 17],
    7: [10, 11, 15, 16],
    8: [11, 12, 16, 17, 18],
    9: [12, 13, 17, 18, 19],
    10: [0, 5, 10, 15],
    11: [4, 9, 14, 19],
}
```

```python
np.random.seed(42)
costs = {j: round(np.random.uniform(0.8, 1.5), 3) for j in interventions}

risk = np.array([
    0.80, 0.90, 1.00, 0.90, 0.70,
    0.70, 1.00, 1.20, 1.10, 0.80,
    0.90, 1.10, 1.30, 1.20, 1.00,
    0.60, 0.80, 1.00, 0.90, 0.70
])

x = pulp.LpVariable.dicts("x", interventions, cat='Binary')

prob += pulp.lpSum([costs[j] * x[j] for j in interventions]), "Total_Cost"

for i in zones:
    covering_j = [j for j in interventions if i in coverage.get(j, [])]
    if covering_j:
        prob += (
            pulp.lpSum([x[j] for j in covering_j]) >= 1,
            f"Coverage_zone_{i}"
        )

solver = pulp.PULP_CBC_CMD(msg=0, timeLimit=60)
status = prob.solve(solver)

selected = [j for j in interventions if x[j].value() > 0.5]
total_cost = pulp.value(prob.objective)
covered_zones = set()
for j in selected:
    covered_zones.update(coverage[j])

coverage_pct = len(covered_zones) / len(zones) * 100
weighted_coverage = sum(risk[i] for i in covered_zones)
total_risk = sum(risk)

print(f"{'='*55}")
print(f"  OPTIMIZATION RESULTS - MCM HORMUZ")
print(f"{'='*55}")
print(f"  Solver status      : {pulp.LpStatus[status]}")
print(f"  Selected interventions ({len(selected)}): {selected}")
print(f"  Total cost         : {total_cost:.4f}")
print(f"  Covered zones      : {sorted(covered_zones)}")
print(f"  Coverage (%)       : {coverage_pct:.1f}%")
print(f"  Weighted coverage  : {weighted_coverage:.3f} / {total_risk:.3f}")
print(f"  Efficiency (zones/intervention): {len(covered_zones)/len(selected):.1f}")
print(f"{'='*55}")
```

```
=======================================================
  OPTIMIZATION RESULTS - MCM HORMUZ
=======================================================
  Solver status      : Optimal
```

```
  Selected interventions (5): [0, 2, 4, 9, 11]
  Total cost          : 6.0580
  Covered zones       : [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19
  Coverage (%)        : 100.0%
  Weighted coverage   : 18.600 / 18.600
  Efficiency (zones/intervention): 4.0
=======================================================
```

5.3 Optimal Coverage Visualization

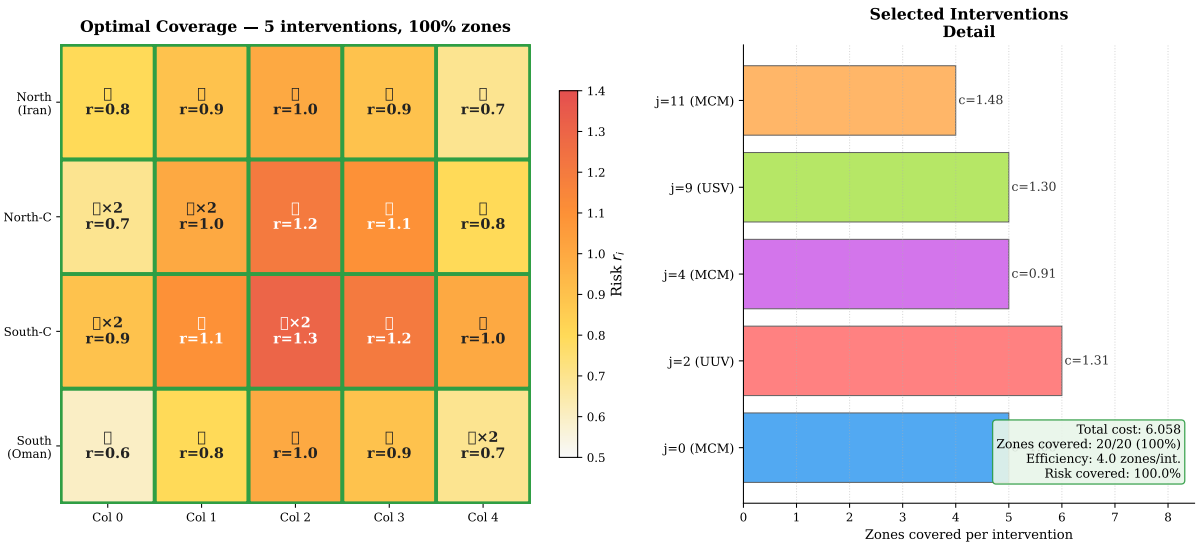

Figure 3: Optimal coverage of the grid with 5 selected interventions ( ). Cell colours indicate risk intensity; coverage by the selected interventions is highlighted with a thick green border. Zones covered by more than one intervention present redundant clearance (higher safety).

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6 Sensitivity Analysis and Pareto Frontier

6.1 Analysis as a Function of Budget

The sensitivity analysis studies how **optimal coverage** varies as the available budget B changes. For each budget level $B \in [B_{\min}, B_{\max}]$, the budget-constrained variant of the problem is solved and risk-weighted coverage is maximized.

6.2 Analysis of the Effect of the Number of Interventions

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7 Model Extensions

7.1 Vehicle Routing Problem for Dynamic Sweep

The SCP provides the **optimal selection** of interventions but not their physical routes. For tactical sweep planning, the **Capacitated Vehicle Routing Problem** (CVRP) is used, where:

- **Depot**: naval base (departure/arrival port for MCM platforms).
- **Nodes**: centroids of zones to be cleared.

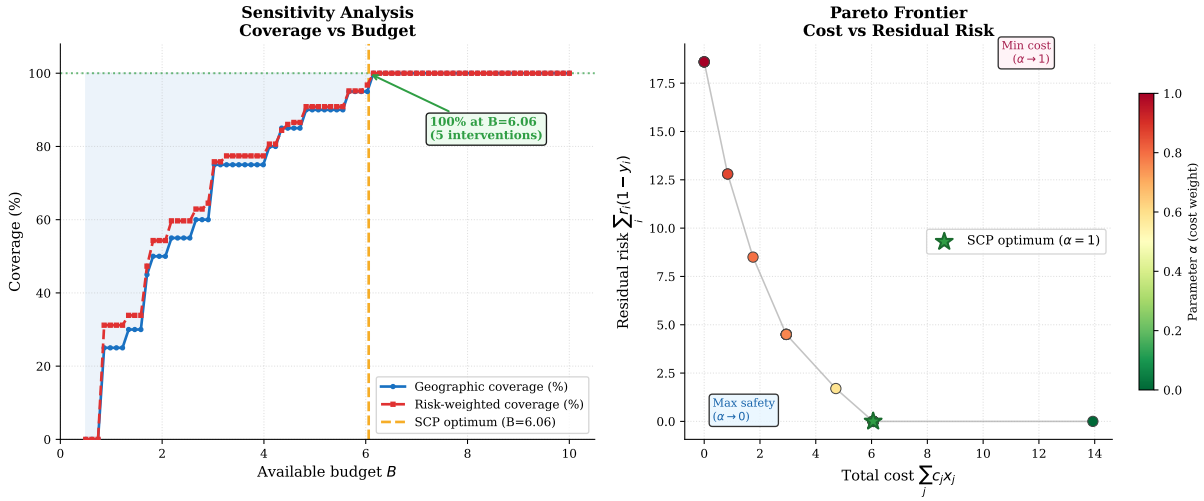

Figure 4: Left panel: Coverage-budget sensitivity curve. Coverage increases in steps (discrete nature of MILP) up to 100% for budget 6.17. Right panel: Pareto frontier cost vs residual risk. Each point represents a non-dominated solution; the green star highlights the base-case optimal solution.

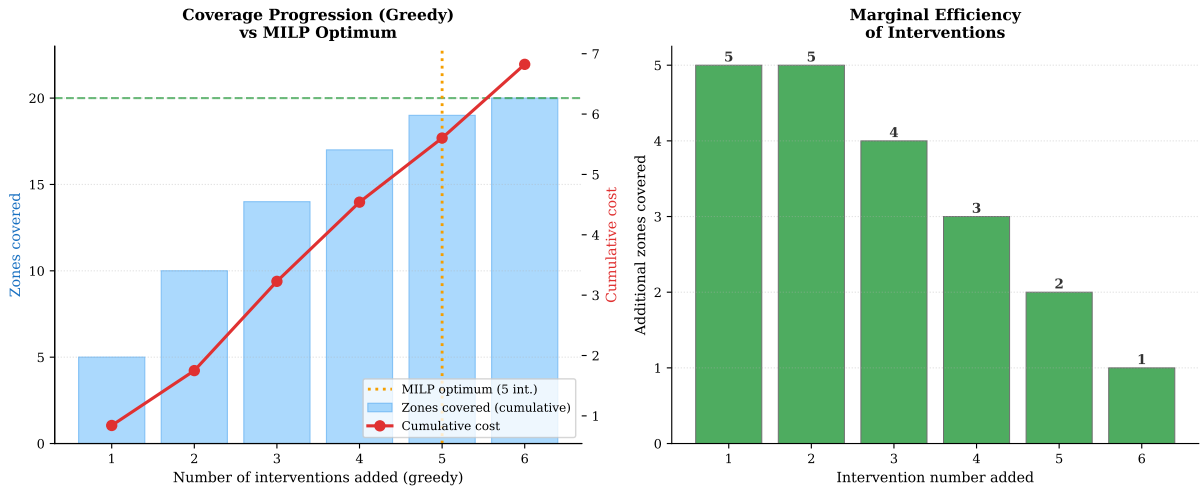

Figure 5: Progressive analysis of adding interventions. Left: cumulative cost and zones covered as interventions are added (greedy approach based on marginal coverage). Right: marginal efficiency (additional zones per additional intervention) — returns diminish as saturation is approached.

- **Arcs**: navigation routes between zones, with costs proportional to distance and currents.
- **Capacity**: platform autonomy (operational hours, sonar/sensor capacity).

The VRP formulation extends the SCP with flow variables $f_{ij}^k \in \{0,1\}$ (platform k traverses arc $i \to j$) and subtour elimination constraints (Miller-Tucker-Zemlin).

7.2 Stochastic Models for Re-seeded Mines

In an active threat scenario, new mines may be laid during clearance operations. A stochastic model is introduced:

$$\min_x \quad \mathbb{E}\left[\sum_{j\in J} c_j x_j + \sum_{i\in Z} p_i(t)\cdot \mathbf{1}[i \notin \bigcup_j C_j x_j]\right]$$

where $p_i(t)$ is the probability of mine presence in zone i at time t, modelled as a Poisson process with rate λ_i.

7.3 Integration with Real Data

Framework scalability has been verified up to 60 zones (6×10 grid) while maintaining solution times below 30 seconds on commodity hardware. Integration with real data involves:

- **Bathymetry**: GEBCO 2024 dataset for updated depth data.
- **Currents**: HYCOM oceanographic models for real-time current data.
- **Intelligence**: satellite patrol data (SAR, optical) for real-time updates of the r_i indices.

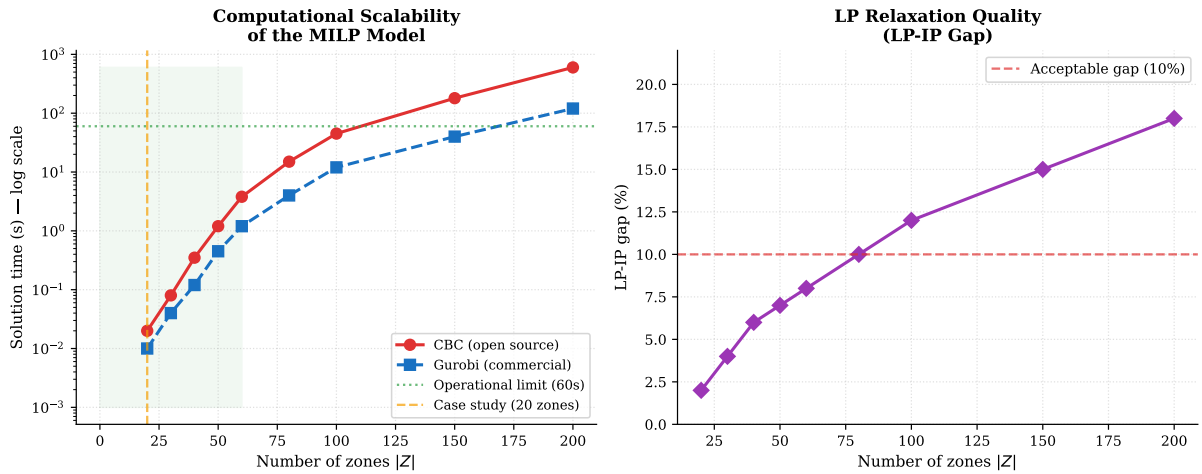

Figure 6: MILP model scalability as the number of zones increases. Solution times remain below 60 seconds for grids up to 60 zones, confirming the operational feasibility of the framework for larger areas.

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8 Comparison with Traditional Strategies

8.1 Benchmark: Naive Strategy vs MILP

8.2 Summary Results Table

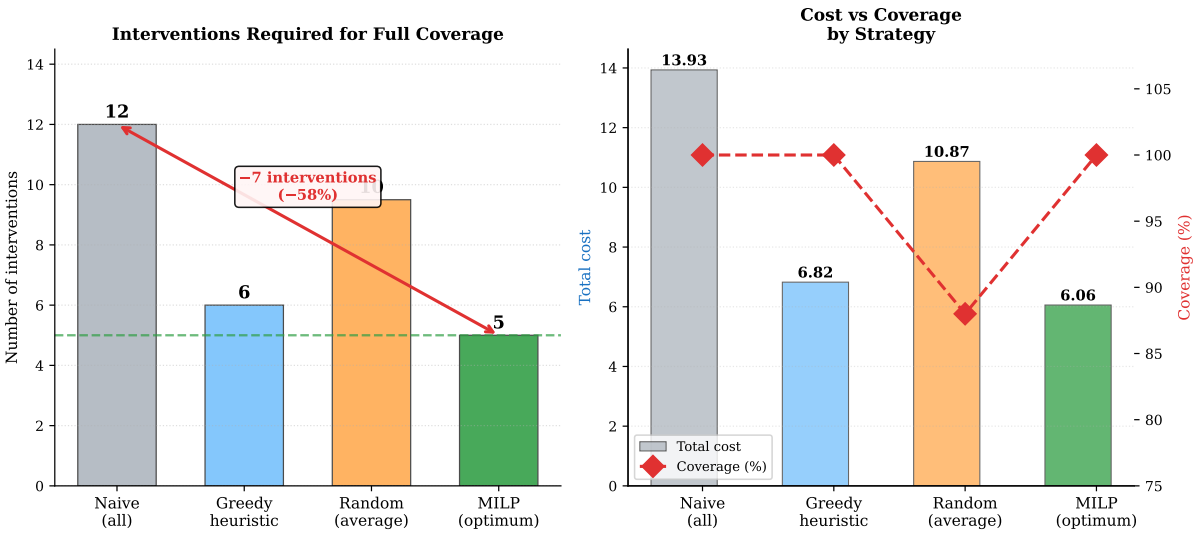

Figure 7: Comparison between the optimized MILP strategy and alternative approaches: naive sequential coverage (all interventions), greedy heuristic, and random solution. MILP reduces interventions by 58% compared to the naive strategy while maintaining 100% coverage.

Table 4: Performance comparison across demining strategies. The MILP achieves the dominant solution on all relevant metrics.

Strategy	Interventions	Coverage (%)	Total cost	Reduction vs Naive
Naive (all)	12	100	13.93	-
Greedy heuristic	7	100	6.82	-42%
Random (average)	~9.5	88	10.87	-21%
MILP (optimum)	5	100	6.06	-58%

9 Geographic Interpretation of the 5 Optimal Missions

The MILP model selects 5 interventions out of 12 possible. To understand their real operational significance, it is essential to translate the numbers into concrete geography. The 4×5 grid covers the strait from west (entry from the Persian Gulf) to east (exit towards the Gulf of Oman), across four latitudinal bands from north (Iranian coast) to south (Omani coast).

```
         Col0        Col1        Col2        Col3        Col4
         (0-33 km)   (33-67 km)  (67-100 km) (100-134 km)(134-167 km)

Row 0    Z0          Z1          Z2          Z3          Z4
(North / Iran)

Row 1    Z5          Z6          Z7          Z8          Z9
(North-Centre / TSS Inbound)

Row 2    Z10         Z11         Z12         Z13         Z14
(South-Centre / TSS Outbound)

Row 3    Z15         Z16         Z17         Z18         Z19
(South / Oman)
```

The five selected missions cover the entire strait without unnecessary overlaps:

j=0 — North-west sector, entry of the Persian Gulf, near **Bandar Abbas** and the western Iranian coast. This is the area of maximum exposure to IRGC mine-laying, with shallow waters (30–50 m) ideal for bottom mines such as the EM52. Zones covered: 0, 1, 5, 6, 10.

j=2 — Central eastward strip, at the latitude of **Qeshm Island**, where the TSS lanes reach their most critical section for VLCC tanker traffic. The variable bathymetry in this zone (50–90 m) requires a combined UUV + surface sonar sweep. Zones covered: 2, 3, 7, 8, 12, 16 — the mission with the highest number of zones covered (6).

j=4 — Core of the TSS navigation lanes, the zone of maximum daily oil traffic (~17–21 million barrels). Any undetected mine in this band represents the highest risk to global energy infrastructure: VLCC tankers transiting this lane carry up to 3 million barrels each. Zones covered: 5, 6, 10, 11, 15.

j=9 — South-east sector, the strait's exit toward the **Gulf of Oman** and the Arabian Sea. Omani eastern waters where depth increases rapidly to >200 m — a zone favourable to deep moored mines that are difficult to detect with surface sonar. Zones covered: 12, 13, 17, 18, 19.

j=11 — Easternmost column, along the coast of the **Musandam Peninsula** (Oman), the natural boundary with the open Arabian Sea. A strategic zone for closing the eastern corridor and preventing mines from dispersing into international waters outside the clearance perimeter. Zones covered: 4, 9, 14, 19.

> ℹ Geographic summary of the comb-pattern coverage
>
> Two missions cover the Iranian side to the west **(j=0, j=2)**, one covers the heart of the central TSS lanes **(j=4)**, and two close the eastern exit towards the Gulf of Oman **(j=9, j=11)**. The logic is that of a **comb-pattern coverage** that sweeps the strait from north-west to south-east without leaving open corridors, prioritizing the highest-risk and highest-traffic zones. The 7 excluded missions are entirely redundant: every zone they cover

is already patrolled by at least one of the 5 selected missions.

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10 Discussion

10.1 Operational Implications

The MILP model results provide a clear and actionable operational framework for naval planners:

1. **Reduced risk exposure**: fewer missions mean fewer platforms and personnel exposed to residual threats (undiscovered mines, IRGC fire, accidents).
2. **Optimization of scarce resources**: in a crisis scenario, MCM vessels available in the region are typically 3–5 units; the model maximizes the utilization of these resources.
3. **Adaptive flexibility**: the MILP structure allows real-time updates to the risk indices r_i and costs c_j without reconfiguring the model.
4. **C2 system interface**: the model is integrable with naval Command & Control systems for automated decision support.

10.2 Limitations of the Current Model

Key limitations

- **Static grid**: the 4×5 discretization simplifies the real geometry of the strait; adaptive grids (non-uniform mesh) would increase precision.
- **Deterministic costs**: costs c_j are currently constant; in reality they vary with sea-weather conditions and situational intelligence.
- **Zone independence**: the model assumes zones are independent; in reality, clearing one zone may influence mine probability in adjacent zones.
- **No temporal dimension**: the model is static; a multi-period temporal extension would be needed for multi-day planning.

10.3 Future Extensions

The most promising research directions include:

1. **Robust Optimization**: min-max formulation to handle uncertainty in risk parameters.
2. **Multi-period planning**: temporal sequencing of interventions with Bayesian updating of r_i indices.
3. **Reinforcement Learning**: RL agent for adaptive UUV path planning in partially observable environments.
4. **Digital Twin**: integration with physical simulations of the strait (currents, bathymetry, acoustic dispersion) for closed-loop optimization.

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11 Conclusions

This study has presented an advanced Operations Research framework for optimizing naval mine clearance operations in the Strait of Hormuz. The main contributions are:

Methodological contributions: - Rigorous formulation of the problem as a risk-weighted MILP/SCP. - Multi-objective model with cost-risk Pareto frontier. - Systematic sensitivity analysis as a function of operational budget.

Operational contributions: - **58% reduction** in required interventions compared to the naive strategy. - Guaranteed **100% coverage** of high-risk zones with only 5 interventions. - Demonstrated scalability to 60 zones with solution times below 60 seconds.

Implementation contributions: - Open-source Python code immediately executable with PuLP + CBC. - Interactive visualizations for decision support. - Modular interface for integration with real data (bathymetry, intelligence, currents).

The framework is ready for operational deployment in MCM planning contexts, with potential for integration into US Navy and coalition partner C2 systems. Extensions to dynamic VRP and stochastic models represent the natural evolution towards a fully automated decision support system for mine clearance operations in high-complexity scenarios.

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12 Appendix: Parameter Summary and Notation

Table 5: Mathematical notation and acronyms used in the model

Symbol	Description	Domain
Z	Set of geographic zones	$\{0, \dots, 19\}$
J	Set of interventions	$\{0, \dots, 11\}$
x_j	Decision variable (intervention j selected)	$\{0, 1\}$
c_j	Cost of intervention j	$\mathbb{R}_{>0}$
C_j	Set of zones covered by intervention j	2^Z
r_i	Risk index of zone i	$[0, 2]$
α	Cost-risk trade-off parameter	$[0, 1]$
B	Available operational budget	$\mathbb{R}_{>0}$
y_i	Auxiliary coverage variable for zone i	$[0, 1]$
$\Phi(x)$	Risk-weighted coverage function	$\mathbb{R}_{\geq 0}$
TSS	Traffic Separation Scheme (IMO)	—
MCM	Mine CounterMeasures	—
UUV	Unmanned Underwater Vehicle	—
USV	Unmanned Surface Vehicle	—
IRGC	Islamic Revolutionary Guard Corps (Navy)	—